BanglaAudioNative: Toward an End-to-End, Text-Free, Full-Duplex Speech Model for Bangla

Research Concept Note (Version 0.1)

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Status. Concept note recording a research idea and plan. No dataset has been collected, no model has been trained, and no experimental results exist at the time of writing. All figures below are target objectives, not findings.

Abstract

Traditional conversational AI systems rely on a cascaded pipeline of automatic speech recognition (ASR), a text-based large language model (LLM), and text-to-speech (TTS) synthesis. Although functional, this text-mediated design adds latency at every stage, discards acoustic and emotional cues such as prosody and tone, and handles natural turn-taking poorly, including interruptions and concurrent backchanneling. We propose **BanglaAudioNative**, a research plan to build an end-to-end, audio-to-audio speech architecture for Bangla that requires no intermediate text transcription at inference. The system would model dual-stream conversational audio through a neural audio codec and an auto-regressive multi-stream transformer, mapping Bangla phonetics and prosody directly from speech to speech. To support training, we plan to build a dual-channel conversational speech dataset of spontaneous, multi-dialect Bangladeshi speech, targeting approximately 50 hours, that includes natural overlaps and Bangla–English ("Benglish") code-switching. We will adapt an open-source full-duplex speech framework using parameter-efficient fine-tuning, with target objectives of sub-second response latency and real-time interruption handling. If successful, this work would address a gap in end-to-end voice intelligence for Bangla and provide an open blueprint for South Asian full-duplex speech systems.

Keywords

Bangla speech; full-duplex dialogue; spoken language models; neural audio codec; code-switching; low-resource languages

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